

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Artificial Intelligence

2. **Subject Name:** Signal Processing Foundations for AI and ML  
**L-T-P:**            **3-1-0**            **Credits: 4**

3. **Pre-requisites:** None

### Syllabus and reference books:

#### Syllabus:

|                                          |                                                                                                                                                                                                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                      | Continuous and discrete-time signals and systems. Linear time invariant system and Convolution. Discrete and continuous representations and processing in deep neural networks.                                                            |
|                                          | Representation of Signals via Fourier Transform (Fourier series, Continuous-time and discrete-time Fourier transform, DFT, etc). Fourier Transform properties and applications of transforms in deep neural networks (e.g., Fourier CNNs). |
|                                          | Characterization of Signals and Systems in time and transform domain: Magnitude-phase representation of Fourier transform, Linear and non-linear phase.                                                                                    |
|                                          | Sampling of Continuous-time Signals. Nyquist criterion. Sampling in spatial and temporal domains.                                                                                                                                          |
| <b>Speech Signal Processing</b>          | Acquisition, Sampling, & Representation of Speech. Signal energy, loudness and decibel.                                                                                                                                                    |
|                                          | Time-domain and transform-domain analysis of speech (including STFT and Spectrograms). Cepstrum and Mel-frequency Cepstral Coefficients.                                                                                                   |
|                                          | Basic speech processing techniques (e.g., noise attenuation and echo cancellation): Traditional and deep learning-based techniques.                                                                                                        |
|                                          | Deep Neural Networks for Speech Signal Processing (e.g., speech recognition, localization and tracking).                                                                                                                                   |
| <b>Image and Video Signal Processing</b> | Acquisition (pinhole and lens-based models) Representations, and basic manipulations of Images and Videos.                                                                                                                                 |
|                                          | Two-dimensional systems and mathematical preliminaries: Linear systems, shift invariance, block matrices and Kronecker products. Image transforms. Speeding up CNNs using 2D-FFT.                                                          |
|                                          | Time-domain and transform-domain analysis of images: Total variation of natural images, Phase-dominance of Natural Images, Energy compaction property of DCT.                                                                              |
|                                          | Basic Image and video processing techniques: Inverse and Wiener filtering. Basics of Ill-posedness, regularization, and                                                                                                                    |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | estimators. Deep Wiener deconvolution: Wiener meets Deep Learning. PCA: PCA meets Deep Learning.                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Deep Neural Networks for Image and Video processing (e.g., Image classification and Image restoration, Video-based event recognition and summarization).                                                                                                                |
| <b>Complex-valued Signal Processing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Acquisition & Representation of complex-valued signals (wind data, synthetic aperture radar and MRI data)                                                                                                                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optimization of complex-valued neural networks. Analyticity and Cauchy-Riemann Condition, Wirtinger Calculus. Processing complex-valued signals in AI.                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Deep Neural Networks for Complex-valued Signal processing (e.g., forecasting of wind profile and SAR denoising).                                                                                                                                                        |
| <b>Multi-modal Fusion in AI and Introduction of Signal Processing+AI in diff Domains</b>                                                                                                                                                                                                                                                                                                                                                                                                        | Multimodal fusion: Applications and challenges from fusing two or more modalities such as vision, audio, biomedical signal. Development of shallow and deep networks for multimodal learning.                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Deep learning based techniques for — Agriculture domain: SAR remote sensing for agriculture applications; Robotics: Stereo vision, Time-of-flight active ranging sensors, Geometric active ranging sensors; Biomedical: Intro to biomedical signals and their analysis. |
| <b>Reference Books:</b> <ol style="list-style-type: none"> <li>1) Signals and Systems, Alan V. Oppenheim, Alan S. Willsky, Prentice Hall, second edition.</li> <li>2) Digital Processing of Speech Signals, L. Rabiner and R. W. Schafer, Prentice Hall, first edition.</li> <li>3) Fundamentals of Digital Image Processing, Anil K Jain, Pearson publishers, first edition.</li> <li>4) Digital Image Processing, Rafael C. Gonzalez and Woods, Pearson publishers, third edition.</li> </ol> |                                                                                                                                                                                                                                                                         |

### 5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                             | No. of lectures | No: of Tutorials |
|-----------------------|-----------------------------------------------------------------------------------|-----------------|------------------|
| 1.                    | Introduction to Signals and Systems                                               | 13              | 2                |
| 2.                    | Speech Signal Processing                                                          | 8               | 2                |
| 3.                    | Image and Video Signal Processing                                                 | 11              | 2                |
| 4.                    | Complex-valued Signal Processing                                                  | 3               | 1                |
| 5.                    | Multi-modal Fusion in AI and Introduction of Signal Processing+AI in diff Domains | 4               | 2                |
| Total number of hours |                                                                                   | 39              | 9 = 48 hours     |